

Q1 - 24 January - Shift 1

Let $y = y(x)$ be the solution of the differential equation $x^3 dy + (xy - 1) dx = 0$, $x > 0$,

$y\left(\frac{1}{2}\right) = 3 - e$. Then $y(1)$ is equal to

- (1) 1
- (2) e
- (3) $2 - e$
- (4) 3

Space for your notes:

Q2 - 24 January - Shift 2

Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)dx + 3xy dy = 0$, $y(1) = 1$. Then

$6y^2(e)$ is equal to

- (1) $3e^2$
- (2) e^2
- (3) $2e^2$
- (4) $\frac{3e^2}{2}$

Space for your notes:

Q3 - 25 January - Shift 1

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}(1 + xy^2(1 + \log_e x))$,

$x > 0, y(1) = 3$. Then $\frac{y^2(x)}{9}$ is equal to :

(1) $\frac{x^2}{5 - 2x^3(2 + \log_e x^3)}$

(2) $\frac{x^2}{2x^3(2 + \log_e x^3) - 3}$

(3) $\frac{x^2}{3x^3(1 + \log_e x^2) - 2}$

(4) $\frac{x^2}{7 - 3x^3(2 + \log_e x^2)}$

Space for your notes:

Q4 - 25 January - Shift 2

Let $y = y(t)$ be a solution of the differential equation

$$\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$$

Where , $\alpha > 0, \beta > 0$ and $\gamma > 0$. Then $\lim_{t \rightarrow \infty} y(t)$

(1) is 0 (2) does not exist

(3) is 1 (4) is -1

Space for your notes:

Q5 - 29 January - Shift 1

Let $y = f(x)$ be the solution of the differential equation $y(x+1) dx - x^2 dy = 0$, $y(1) = e$. Then $\lim_{x \rightarrow 0^+} f(x)$ is equal to

Space for your notes:

(1) 0

(2) $\frac{1}{e}$

(3) e^2

(4) $\frac{1}{e^2}$

Q6 - 29 January - Shift 2

Let $y = y(x)$ be the solution of the differential equation $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x$, ($x > 1$). If $y(2) = 2$, then $y(e)$ is equal to

Space for your notes:

(1) $\frac{4+e^2}{4}$

(2) $\frac{1+e^2}{4}$

(3) $\frac{2+e^2}{2}$

(4) $\frac{1+e^2}{2}$

Q7 - 30 January - Shift 1

Let the solution curve $y = y(x)$ of the differential equation

$$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{\frac{3}{2}}} y = 2x$$

$\exp \frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x)^6}}$ pass through the origin. Then

$y(1)$ is equal to :

- (1) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$ (2) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$
 (3) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (4) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$

Space for your notes:

Q8 - 30 January - Shift 2

The solution of the differential equation

$$\frac{dy}{dx} = -\left(\frac{x^2 + 3y^2}{3x^2 + y^2}\right), y(1) = 0 \text{ is}$$

- (1) $\log_e |x+y| - \frac{xy}{(x+y)^2} = 0$
 (2) $\log_e |x+y| + \frac{xy}{(x+y)^2} = 0$
 (3) $\log_e |x+y| + \frac{2xy}{(x+y)^2} = 0$
 (4) $\log_e |x+y| - \frac{2xy}{(x+y)^2} = 0$

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Q9 - 31 January - Shift 2

Let $y = y(x)$ be the solution of the differential equation

$$(3y^2 - 5x^2)y \, dx + 2x(x^2 - y^2) \, dy = 0 \text{ such}$$

that $y(1) = 1$. then $\left| (y(2))^3 - 12y(2) \right|$ is equal

to:

(1) $32\sqrt{2}$

(2) 64

(3) $16\sqrt{2}$

(4) 32

Space for your notes:

Q10 - 01 February - Shift 1

If $y = y(x)$ is the solution curve of the differential equation $\frac{dy}{dx} + y \tan x = x \sec x$, $0 \leq x \leq \frac{\pi}{3}$,

$y(0) = 1$, then $y\left(\frac{\pi}{6}\right)$ is equal to

(1) $\frac{\pi}{12} - \frac{\sqrt{3}}{2} \log_e \left(\frac{2}{e\sqrt{3}} \right)$

(2) $\frac{\pi}{12} + \frac{\sqrt{3}}{2} \log_e \left(\frac{2\sqrt{3}}{e} \right)$

(3) $\frac{\pi}{12} - \frac{\sqrt{3}}{2} \log_e \left(\frac{2\sqrt{3}}{e} \right)$

(4) $\frac{\pi}{12} + \frac{\sqrt{3}}{2} \log_e \left(\frac{2}{e\sqrt{3}} \right)$

Space for your notes:

Q11 - 01 February - Shift 2

Let $\alpha x = \exp(x^\beta y^\gamma)$ be the solution of the differential equation $2x^2y \, dy - (1 - xy^2) \, dx = 0$,

$x > 0, y(2) = \sqrt{\log_e 2}$. Then $\alpha + \beta - \gamma$ equals :

- (1) 1
- (2) -1
- (3) 0
- (4) 3

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1)

Q2 (3)

Q3 (1)

Q4 (1)

Q5 (1)

Q6 (1)

Q7 (1)

Q8 (3)

Q9 (1)

Q10 (1)

Q11 (1)

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Q1 (1)

$$\frac{dy}{dx} = \frac{1-xy}{x^3} = \frac{1}{x^3} - \frac{y}{x^2}$$

$$\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^3}$$

$$\text{If } = e^{\int \frac{1}{x^2} dx} = e^{-\frac{1}{x}}$$

$$y \cdot e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}} \cdot \frac{1}{x^3} dx \quad \left(\text{put } -\frac{1}{x} = t \right)$$

$$y \cdot e^{-\frac{1}{x}} = -\int e^t \cdot t dt$$

$$y = \frac{1}{x} + 1 + Ce^{\frac{1}{x}}$$

Where C is constant

$$\text{Put } x = \frac{1}{2}$$

$$3 - e = 2 + 1 + Ce^2$$

$$C = -\frac{1}{e}$$

$$y(1) = 1$$

Q2 (3)

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$$(x^2 - 3y^2) dx + 3xy dy = 0$$

$$\frac{dy}{dx} = \frac{3y^2 - x^2}{3xy} \Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{3} \frac{x}{y} \quad (1)$$

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$(1) \Rightarrow v + x \frac{dv}{dx} = v - \frac{1}{3} \frac{1}{v}$$

$$\Rightarrow v dv = \frac{-1}{3x}$$

Integrating both side

$$\frac{v^2}{2} = \frac{-1}{3} \ln x + c$$

$$\Rightarrow \frac{y^2}{2x^2} = \frac{-1}{3} \ln x + c$$

$$y(1) = 1$$

$$\Rightarrow \boxed{\frac{1}{2} = c}$$

$$\Rightarrow \frac{y^2}{2x^2} = \frac{-1}{3} \ln x + \frac{1}{2}$$

$$\Rightarrow y^2 = x^2 \left(-\frac{2}{3} \ln x + 1 \right)$$

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Q3 (1) $y^2(e) = -\frac{2}{3}e^2 + e^2 = \frac{e^2}{3}$

$$\frac{dy}{dx} - \frac{y}{x} = y^3(1 + \log_e x)$$

$$\frac{1}{y^3} \frac{dy}{dx} - \frac{1}{xy^2} = 1 + \log_e x$$

$$\text{Let } -\frac{1}{y^2} = t \Rightarrow \frac{2}{y^3} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \frac{dt}{dx} + \frac{2t}{x} = 2(1 + \log_e x)$$

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = x^2$$

$$\frac{-x^2}{y^2} = \frac{2}{3} \left((1 + \log_e x) x^3 - \frac{x^3}{3} \right) + C$$

$$y(1) = 3$$

$$\frac{y^2}{9} = \frac{x^2}{5 - 2x^3(2 + \log_e x^3)}$$

Q4 (1)

$$\frac{dy}{dt} + \alpha y = \gamma e^{-\beta t}$$

$$\text{I.F.} = e^{\int \alpha dt} = e^{\alpha t}$$

$$\text{Solution} \Rightarrow y \cdot e^{\alpha t} = \int \gamma e^{-\beta t} \cdot e^{\alpha t} dt$$

$$\Rightarrow y e^{\alpha t} = \gamma \frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)} + c$$

$$\Rightarrow y = \frac{\gamma}{e^{\beta t}(\alpha-\beta)} + \frac{c}{e^{\alpha t}}$$

$$\text{So, } \lim_{t \rightarrow \infty} y(t) = \frac{\gamma}{\infty} + \frac{c}{\infty} = 0$$

Q5 (1)

$$\frac{x+1}{x^2} dx = \frac{dy}{y}$$

$$\ln x - \frac{1}{x} = \ln y + c$$

$$(1, e)$$

$$c = -2$$

$$\ln x - \frac{1}{x} = \ln y - 2$$

$$y = e^{\ln x} - \frac{1}{x} + 2$$

$$\lim_{x \rightarrow 0^+} e^{\ln x - 1} - \frac{1}{x} + 2$$

$$= e^{-\infty}$$

$$= 0$$

Q6 (1)

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$$x \log_e x \frac{dy}{dx} + y = x^2 \log_e x, (x > 1).$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x \ln x} = x$$

Linear differential equation

$$\text{I.F.} = e^{\int \frac{1}{x \ln x} dx} = |\ln x|$$

$\therefore$ Solution of differential equation

$$y|\ln x| = \int x|\ln x| dx$$

$$= |\ln x| \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx$$

$$\Rightarrow y|\ln x| = |\ln x| \left(\frac{x^2}{2} \right) - \frac{x^2}{4} + c$$

For constant

$$y(2) = 2 \Rightarrow c = 1$$

$$\text{So, } y(x) = \frac{x^2}{2} - \frac{x^2}{4|\ln x|} + \frac{1}{|\ln x|}$$

$$\text{Hence, } y(e) = \frac{e^2}{2} - \frac{e^2}{4} + 1 = 1 + \frac{e^2}{4}$$

Q7 (1)

$$\frac{dy}{dx} + \left(\frac{-3x^5 \tan^{-1} x^3}{(1+x^6)^{3/2}} \right) y = 2e^{\left\{ \frac{x - \tan x}{\sqrt{1+x^6}} \right\}}$$

$$\text{I.F.} = e^{\int \frac{-3x^5 \tan^{-1} x^3}{(1+x^6)^{3/2}} dx}$$

$$= e^{\frac{\tan^{-1} x^3 - x^3}{\sqrt{1+x^6}}}$$

Solution of differential equation

$$y \cdot e^{\frac{\tan^{-1} x^3 - x^3}{\sqrt{1+x^6}}} = \int 2x e^{\left(\frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}} \right)} \cdot e^{\left(\frac{\tan^{-1} (x^3) - x^3}{\sqrt{1+x^6}} \right)} dx$$

$$= \int 2x dx + c$$

$$y \cdot e^{\frac{\tan^{-1} x^3 - x^3}{\sqrt{1+x^6}}} = x^2 + c$$

Also it passes through origin

$$c = 0$$

$$y(1) \cdot e^{\frac{\tan^{-1}(1) - 1}{\sqrt{2}}} = 1$$

$$y(1) \cdot e^{\frac{\frac{\pi}{4} - 1}{\sqrt{2}}} = 1$$

$$y(1) \cdot e^{\frac{\pi - 4}{4\sqrt{2}}} = 1$$

$$y(1) = \frac{1}{e^{\frac{\pi - 4}{4\sqrt{2}}}} = e^{\frac{4 - \pi}{4\sqrt{2}}}$$

Q8 (3)

Put $y = vx$

$$v + x \frac{dv}{dx} = - \left(\frac{1+3v^2}{3+v^2} \right)$$

$$x \frac{dv}{dx} = - \frac{(v+1)^3}{3+v^2}$$

$$\frac{(3+v^2)dv}{(v+1)^3} + \frac{dx}{x} = 0$$

$$\int \frac{4dv}{(v+1)^3} + \int \frac{dv}{v+1} - \int \frac{2dv}{(v+1)^2} + \int \frac{dx}{x} = 0$$

$$\frac{-2}{(v+1)^2} + \ln(v+1) + \frac{2}{v+1} + \ln x = c$$

$$\frac{-2x^2}{(x+y)^2} + \ln\left(\frac{x+y}{x}\right) + \frac{2x}{x+y} + \ln x = c$$

$$\frac{2xy}{(x+y)^2} + \ln(x+y) = c$$

$\therefore c = 0$, as $x = 1, y = 0$

$$\frac{2xy}{(x+y)^2} + \ln(x+y) = 0$$

Q9 (1)

$$(3y^2 - 5x^2)y \cdot dx + 2x(x^2 - y^2)dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(5x^2 - 3y^2)}{2x(x^2 - y^2)}$$

Put $y = mx$

$$\Rightarrow m + x \cdot \frac{dm}{dx} = \frac{m(5 - 3m^2)}{2(1 - m^2)}$$

$$x \cdot \frac{dm}{dx} = \frac{(5 - 3m^2)m - 2m(1 - m^2)}{2(1 - m^2)}$$

$$\Rightarrow \frac{dx}{x} = \frac{2(m^2 - 1)}{m(m^2 - 3)} dm$$

$$\Rightarrow \frac{dx}{x} = \left(\frac{2}{m} - \frac{\frac{4}{3}}{m} + \frac{\frac{4m}{3}}{m^2 - 3} \right) dm$$

$$\Rightarrow \int \frac{dx}{x} = \int \frac{\left(\frac{2}{3}\right)}{m} + \int \frac{2}{3} \left(\frac{2m}{m^2 - 3} \right) dm$$

$$\Rightarrow \ln |x| = \frac{2}{3} \ln |m| + \frac{2}{3} \ln |m^2 - 3| + C$$

$$\text{Or, } \ln |x| = \frac{2}{3} \ln \left| \frac{y}{x} \right| + \frac{2}{3} \ln \left| \left(\frac{y}{x} \right)^2 - 3 \right| + C$$

$$\text{Put } (x=1, y=1): \text{ we get } c = -\frac{2}{3} \ln(2)$$

$$\Rightarrow \ln |x| = \frac{2}{3} \ln \left| \frac{y}{x} \right| + \frac{2}{3} \ln \left| \left(\frac{y}{x} \right)^2 - 3 \right| - \frac{2}{3} \ln(2)$$

$$\Rightarrow \left(\frac{y}{x} \right) \left[\left(\frac{y}{x} \right)^2 - 3 \right] = 2 \cdot (x^{3/2})$$

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Questions (with Solutions) $4 \times 2 \times 2 \times 2 \sqrt{2}$

MathonGo

Q10 (1) $y^3 - 12y = 32\sqrt{2}$

Here I.F. = $\sec x$

Then solution of D.E :

$$y(\sec x) = x \tan x - \ln(\sec x) + c$$

$$\text{Given } y(0) = 1 \Rightarrow c = 1$$

$$\therefore y(\sec x) = x \tan x - \ln(\sec x) + 1$$

$$\text{At } x = \frac{\pi}{6}, y = \frac{\pi}{12} + \frac{\sqrt{3}}{2} \ln \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$$

Q11 (1)

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$$\alpha x = e^{x^\beta \cdot y^\gamma}$$

$$2x^2 y \frac{dy}{dx} = 1 - x \cdot y^2$$

$$y^2 = t$$

$$x^2 \frac{dt}{dx} = 1 - xt$$

$$\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\text{I.F.} = e^{\int \frac{1}{x} dx} = x$$

$$t(x) = \int \frac{1}{x^2} \cdot x \, dx$$

$$y^2 \cdot x = \ln x + C$$

$$\therefore 2 \cdot \ln 2 = \ln 2 + C$$

$$\therefore C = \ln 2$$

$$\text{Hence, } xy^2 = \ln 2x$$

$$\therefore 2x = e^{x \cdot y^2}$$

$$\text{Hence } \alpha = 2, \beta = 1, \gamma = 2$$